Opening Up Official Statistics in R with pxweb

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Abstract

The **pxweb** package provides an interface from R to statistical databases published through the PxWeb API. PxWeb is used by national, regional, and international statistical organizations to disseminate official statistics in multidimensional tables. The package supports discovery of API resources, construction and validation of table queries, retrieval and parsing of data and metadata, conversion to standard R data structures, extraction of comments, and citation of retrieved data. This article introduces the motivation and design of **pxweb**, outlines the support for both PxWeb API version 1 and version 2, and provides a reproducible case study using official statistics.

Keywords: open data, citing data, public data, pxweb, PC-Axis, R.

1. Introduction

Open data science workflows rely heavily on algorithmic tools for data retrieval. Automating major parts of the data workflow, such as finding, accessing, integrating, citing, and reporting data, helps users spend more time on statistical analysis and interpretation. It also improves reproducible research (Gandrud 2013; Boettiger, Chamberlain, Hart, and Ram 2015) by making data acquisition both explicit and repeatable.

Official statistics are a particularly important class of public data (United Nations General Assembly 2014). National statistical institutes and international agencies publish large collections of data on demography, economics, health, infrastructure, climate, and other areas (United Nations General Assembly 2014; World Bank 2026). Such data are often multidimensional, geographically detailed, and updated over long time periods. Opening these resources through public web interfaces is, however, only the first step. Users also need software that can bridge the gap between online statistical databases and reproducible analysis in statistical programming environments such as R.

Dedicated software packages simplify, standardize, and automate analysis workflows. They take into account variations in raw data formats, access details, metadata conventions, and typical use cases so that users can avoid repetitive programming tasks and reduce the risk of misinterpretation and processing errors (Groves and Lyberg 2010). In R, several packages support programmatic data retrieval from specific sources (Lahti, Huovari, Kainu, and Biecek 2017; Eugster and Schlesinger 2012; Arel-Bundock 2013), broader compiled sources such as **pdfetch**, which provides access to economic and financial time-series data (Reinhart 2015), or more generic data types (Szöcs, Stirling, Scott, Scharmüller, and Schäfer 2020; Caro and Biecek 2017). The **rsdmx** package (Blondel 2014) supports the Statistical Data and Metadata eXchange (SDMX) standard, an alternative to PxWeb.

The **pxweb** package fills a complementary role: it provides a general R interface to databases exposed through the PxWeb application programming interface (API) (Statistics Sweden and Statistics Norway 2026). The package can navigate PxWeb resources, fetch metadata, construct and validate queries, retrieve data, convert responses into standard R data structures, and produce citation and provenance information for the source data. As of version 1.0.0, the **pxweb** package includes 46 API catalogue entries and supports both PxWeb API version 1 and version 2.

Data citation is another motivation for the package. Data reuse should document source organizations, accessed resources, version information, and retrieval times where possible but this faces many practical challenges (Parsons, Duerr, and Jones 2019). The **pxweb** package helps standardize and streamline this process for PxWeb data by collecting source information from the API and data objects and by encouraging users to cite both the retrieved data and the software used to retrieve them.

The package was first released on CRAN in 2014 and was substantially upgraded in 2018 and 2026. Recent development has added support for the PxWeb API version 2, including table search, JSON-stat2 responses, v2 metadata, codelists, value sets, aggregation helpers, and observation status comments. The current package is part of the rOpenGov open data science project (Lahti, Parkkinen, Lehtomäki, and Kainu 2013).

1.1. PxWeb and PC-Axis

PxWeb is a web application and API for disseminating PC-Axis statistical data (Statistics Sweden and Statistics Norway 2026). PC-Axis data are typically organized as multidimensional tables with dimensions such as region, time, sex, age, indicator, or content variable. The PX software family is used by statistical producers, e.g., Statistics Sweden, Statistics Finland, and Statistics Norway (Statistics Sweden 2026; Statistics Finland 2026; Statistics Norway 2026). PxWeb exposes these data and their metadata through hypertext transfer protocol (HTTP) endpoints that can be explored interactively or accessed programmatically.

The PX/PC-Axis ecosystem predates current web PxWeb APIs. The PX file-format specification documents a metadata-and-data file structure and records core keywords such as `CONTENTS`, `DATA`, `MATRIX`, `STUB`, and `CODES` with introduction years going back to 1990 (Likidis 2013). Statistics Sweden describes PX as a standard format for statistics files used by many statistical offices and as the basis for tools such as PxWin, PxWeb, and PxEdit (Statistics Sweden 2026). PxWeb extends this file-oriented tradition to online dissemination by publishing multidimensional statistical tables through web interfaces and API endpoints (Statistics Sweden and Statistics Norway 2026). Recent PxWeb API version 2 development, including the Statbank Norway launch in 2025, further modernizes this infrastructure with a more stable URL structure, GET requests, improved filtering, and richer metadata access (Statistics Sweden and Statistics Norway 2026; Statistics Norway 2026).

From the perspective of an R user, PxWeb APIs have three important parts. First, a table query must be constructed from table metadata. Second, the data response has to be retrieved and converted into an analysis-ready format. Third, the downloaded data should be easy to cite to facilitate provenance and reproducibility. The **pxweb** package handles all three parts for the R user.

The package distinguishes between the PxWeb API and the older PC-Axis file format, typically stored as `.px` files (Likidis 2013). The **pxweb** package is designed for PxWeb APIs,

Software type	R packages	Relation to **pxweb**
Provider-specific data clients	**eurostat**, **WDI**, **osmar**	Focus on one data provider or data family rather than the PxWeb API family
Generic data retrieval tools	**pdfetch**, **webchem**, **intsvy**	Retrieve broad data sources but do not target PxWeb metadata and query semantics
PX/PC-Axis file tools	**pxR**	Work with local `.px` files
SDMX-standard tools	**rsdmx**	Acces data via alternative SDMX standard.
Downstream domain packages	**geofi**	Can build specialized workflows on top of official-data access

Table 1: Related software and its relation to **pxweb**.

while packages such as **pxR** focus on parsing legacy PC-Axis files (Bellosta, Viciana, and Lamigueiro 2025). This distinction is important because API access requires metadata discovery, query construction, request validation, and response parsing rather than only local file import.

Because PxWeb installations are configured by individual providers, API version support and request limits are endpoint-level properties rather than global constants. The **pxweb** package therefore collects available provider configuration metadata, including version 1 `?config` responses, and uses this information when validating requests, making calls, and splitting large downloads.

1.2. Related Software

The **pxweb** package sits between general data-retrieval packages and source-specific official-statistics packages. Table 1 summarizes the main distinctions. Whereas **pdfetch** provides access to broad economic and financial time-series sources (Reinhart 2015), other packages support specific data providers such as Eurostat (**eurostat**) (Lahti *et al.* 2017), World Bank (**WDI**) (Arel-Bundock 2013), or Open Street Map (**osmar**) (Eugster and Schlesinger 2012). Furthermore, there are packages supporting generic data types (e.g., **webchem** (Szöcs *et al.* 2020), **intsvy** (Caro and Biecek 2017)) or data standards such as the Statistical Data and Metadata eXchange (SDMX) standard, which is the most widely adopted alternative to PxWeb among statistical agencies (Willekens 2023) supported by the **rsdmx** package (Blondel 2014). In contrast, **pxweb** targets a shared API family used by multiple statistical organizations.

The **pxweb** package targets web APIs, including metadata discovery, query validation, request splitting, and response parsing. File-oriented tools such as **pxR** address a related but different problem: reading and writing local `.px` files (Bellosta *et al.* 2025). The official PxWeb software provides the server-side infrastructure through which statistical producers publish tables and API endpoints (Statistics Sweden and Statistics Norway 2026).

Finally, **pxweb** can serve as infrastructure for more specialized packages and applications. Domain-specific packages can build on its API access, query, and parsing functionality while adding provider-specific analysis, visualization, or educational workflows. For example, the **hetu** (Kantanen, Bülow, Lahtinen, Magnusson, Paananen, and Lahti 2025) and **geofi** packages address complementary parts of the official-data ecosystem by simplifying the handling of personal and organizational identity codes, and access to Finnish geospatial data from Statistics Finland's spatial APIs (Kainu, Lehtomäki, and Lahti 2026), respectively.

Workflow part	Recommended functions and methods
Search and discovery	`pxweb_search()`, `pxweb_interactive()`, `pxweb_api_catalogue()`
Metadata	`pxweb_get()`, `pxweb_codelists()`
Query	`pxweb_query()`, `pxweb_validate_query_with_metadata()`, `pxweb_all()`, `pxweb_latest()`, `pxweb_aggregation()`, `pxweb_valueset()`
Retrieval	`pxweb_get()`, `pxweb_get_data()`
Conversion	`as.data.frame()`
Comments and citation	`pxweb_data_comments()`, `pxweb_cite()`

Table 2: Recommended **pxweb** functions and methods grouped by their main role in the package workflow.

2. The pxweb package

The **pxweb** package is designed as a generic interface rather than a client for one statistical agency or API. Its core design goal is to provide a stable R workflow across PxWeb providers while preserving provider-specific metadata where it affects query construction, codelists, comments, and citation.

In version 1.0.0, the package interface has 39 exported functions, but ordinary analysis workflows are built from a smaller set of recommended functions. The central workflow is built around the `pxweb_get()` function, which retrieves hierarchy levels, table metadata, or table data depending on the supplied URL and query. The helper `pxweb_get_data()` wraps the same workflow and returns a `data.frame`; further utilities (e.g., `as.data.frame()`) are available to integrate the outputs directly with **tidyverse** workflows (Wickham *et al.* 2019) for subsequent filtering, reshaping, and visualization, allowing users to move seamlessly from data retrieval to tidy data analysis (Wickham 2014) and the vast array of advanced statistical and probabilistic programming tools available through the R ecosystem. Query construction is handled by `pxweb_query()` together with helper functions such as `pxweb_all()`, `pxweb_latest()`, and `pxweb_aggregation()`. Table 2 summarizes the main user-facing functions and methods by their role in the recommended workflow.

The package is licensed under the BSD-2-Clause (aka FreeBSD) license, consistent with common practice for research software (Morin, Urban, and Sliz 2012).

Internally, the package represents API URLs, metadata, data, comments, and query objects with S3 classes. This allows common operations such as printing, validation, conversion with `as.data.frame()`, and combining split responses to behave consistently across API versions. The package uses **httr** for HTTP requests and **jsonlite** for JSON parsing (Wickham 2016; Ooms 2014).

The bundled API catalogue is a practical layer on top of the generic API client. It stores provider URL templates, available languages, version paths, aliases, and citation metadata. Users can rely on the catalogue for common providers or pass explicit PxWeb URLs for APIs that are not already bundled with the package. The current catalogue contains 46 entries covering national, regional, and international statistical resources; the complete list is given in Appendix A. At the time of writing, the bundled catalogue entries point mainly to version 1 API paths; version 2 APIs can also be used by passing explicit PxWeb URLs.

A central implementation detail is that user-supplied URLs are first converted into `pxweb` objects. The constructor parses the URL, detects whether the endpoint belongs to PxWeb API version 1 or version 2, retrieves provider configuration metadata, and stores call history in a temporary cache. Provider configuration is normalized into a common internal representation with the number of calls allowed per time window, the length of that time window, and the maximum number of downloadable values or cells. This lets the same retrieval workflow use version 1 `?config` responses and version 2 `/config` responses through a shared interface.

Support for PxWeb API version 2 is expressed in the same top-level workflow as version 1 (Statistics Sweden and Statistics Norway 2026). Version 2 tables use separate metadata and data endpoints and return JSON-stat2 by default. The **pxweb** package detects v2 URLs, builds the corresponding data endpoint, constructs the v2 query body, and adds query parameters for codelists and output values when needed. This keeps the user workflow broadly similar across API versions, even though the underlying endpoints and request details differ.

For data requests, `pxweb_get()` first obtains table metadata, adds mandatory variables when needed, validates the query against the metadata, and computes any required batches before making data requests. Version 1 tables are queried with the table URL and a JSON request body, whereas version 2 tables use the corresponding `/tables/{id}/data` endpoint with version-specific query parameters such as language and output format. The returned pieces are parsed into package-specific S3 objects and combined when the response type supports it.

Large queries can exceed provider-side limits. The batching procedure can be viewed as a constrained multidimensional query-partitioning heuristic. Given the provider's maximum downloadable size, the algorithm first identifies splittable dimensions from table metadata. Time variables can be split, and so can variables marked as eliminable in the provider metadata, meaning that the API allows them to be omitted or partitioned across requests. Content variables and top or bottom selections are kept intact because they affect the meaning or ordering of the returned values. It then evaluates candidate orderings of the splittable dimensions and, for each ordering, greedily chooses the largest dimension chunks that keep each subquery below the provider limit. The ordering with the smallest resulting number of batches is used. For the typical case of fewer than seven splittable dimensions, all orderings are evaluated. Otherwise, a random sample of orderings is used to keep computation bounded.

The package follows common software engineering practices, including version control, automated tests, API fixtures, continuous integration, and test workflows for selected APIs (Perez-Riverol *et al.* 2016). The default local test suite currently uses mocked and offline-safe tests, with live tests controlled by environment variables.

2.1. Functionality

The package supports the full workflow shown in Figure 1: search and discovery, metadata inspection, query construction, retrieval, conversion, comments, and citation. Interactive use starts with `pxweb_interactive()` for version 1, which allows a user to navigate a known API, select a table, build a query, optionally download data, and print reusable `R` code. Programmatic use follows the same steps with explicit function calls.

Installation of the CRAN release version follows the standard procedure in R.

```
install.packages("pxweb")
library("pxweb")
```

Search and discovery start either from the bundled API catalogue, from an explicit URL,

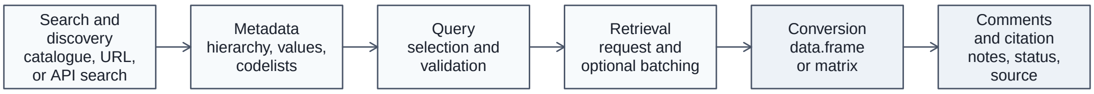

Figure 1: Main **pxweb** workflow: search and discovery, metadata, query construction, retrieval, conversion, comments, and citation.

or, for PxWeb API version 2, from table search. The following example searches Statistics Sweden's PxWeb API version 2 for a population table and stores its metadata URL.

```
search_results <- pxweb_search(
  query = "Population by region marital status age sex",
  api_url = "https://statistikdatabasen.scb.se/api/v2",
  lang = "en",
  page_size = 5
)

url <- search_results$metadata_url[search_results$id == "TAB638"][1]
```

Metadata are retrieved with `pxweb_get()` when no query is supplied. The returned metadata object describes the table dimensions, values, labels, and version-specific metadata needed for query validation and data retrieval. Version 2 APIs may also expose aggregation codelists and value sets, which can be inspected with `pxweb_codelists()`.

```
meta <- pxweb_get(url)
pxweb_codelists(meta, variable = "Alder")[, c("id", "type", "label")]
```

```
               id        type                            label
1 agg_Ålder10årJ Aggregation             ´10-year intervals
2   agg_Ålder5år Aggregation               5-year intervals
3   vs_Ålder1årA    Valueset        Age, 1 year age classes
4   vs_ÅlderTotA    Valueset Age, total, all presented ages
```

The codelist labels are provider-supplied. In this example the English metadata endpoint returns English labels, while the `id` values remain provider identifiers that are passed unchanged to query helpers.

Queries can be written as named R lists and converted to `pxweb_query` objects when needed. For codelists, the `id` column returned by `pxweb_codelists()` is passed to helpers such as pxweb_aggregation() or pxweb_valueset(). Helper functions express common selection patterns such as all values, the latest time period, top or bottom selections, aggregations, and value sets. Queries are validated against metadata before calling the PxWeb API.

```
query <- list(
  Region = "00",
  Civilstand = "OG",
  Alder = pxweb_aggregation("agg_Ålder5år"),
  Kon = c("1", "2"),
  ContentsCode = "BE0101N1",
  Tid = pxweb_latest()
)

px_query <- pxweb_query(query)
```

Retrieval is handled by pxweb_get(). Users who want a data.frame directly can instead use pxweb_get_data(). The same top-level workflow handles version 1 and version 2 URLs, constructs the appropriate request body, and splits large supported queries into batches when needed.

```
px_data <- pxweb_get(url, query = query)
```

Conversion methods turn PxWeb responses into standard R objects. Users can choose whether column names and variable values use provider label or code.

```
px_df <- as.data.frame(
  px_data,
  column.name.type = "code",
  variable.value.type = "code"
)
```

Finally, PxWeb responses may include comments, notes, or status information. The **pxweb** package exposes these through pxweb_data_comments(), with support for both version 1 comments and version 2 dataset notes, dimension notes, value notes, and observation status comments. Citation support is provided by pxweb_cite(), which prints citation information for the retrieved data source and the **pxweb** package itself.

```
comments <- pxweb_data_comments(px_data)
pxweb_cite(px_data)
```

3. Case Study

We illustrate the package with a Statistics Sweden population table that is both familiar official statistics and technically useful for showing large-query handling. The example uses Statistics Sweden's PxWeb API version 2 table search to locate table TAB638, "Population by region, marital status, age and sex. Year 1968–2024". Exact results can change if Statistics Sweden changes table metadata, values, or provider-side configuration.

```
library("pxweb")

search_results <- pxweb_search(
  query = "Population by region marital status age sex",
  api_url = "https://statistikdatabasen.scb.se/api/v2",
  lang = "en",
  page_size = 5
)

metadata_url <- search_results$metadata_url[search_results$id == "TAB638"][1]
```

We use a large but manageable subset of the table: all 21 Swedish counties, all marital-status categories, single-year ages from 0 to 100+, both sexes, the population count content variable, and all years from 1968 to 2024. These codes can all be accessed with the metadata object.

```
px_metadata <- pxweb_get(metadata_url)
```

```
PXWEB METADATA
Population by region, marital status, age and sex.  Year 1968-2024
variables:
 [[1]] Region: region
 [[2]] Civilstand: marital status
 [[3]] Alder: age
 [[4]] Kon: sex
 [[5]] ContentsCode: observations
 [[6]] Tid: year
```

The metadata object contains the labels returned by the provider. With `lang = "en"` in the search call, the selected metadata URL also requests English labels.

```
years <- as.character(1968:2024)
ages <- c(as.character(0:99), "100+")
county_codes <- c(
  "01", "03", "04", "05", "06", "07", "08", "09", "10", "12", "13",
  "14", "17", "18", "19", "20", "21", "22", "23", "24", "25"
)

query <- list(
  Region = county_codes,
  Civilstand = c("OG", "G", "SK", "ÄNKL"),
  Alder = ages,
  Kon = c("1", "2"),
  ContentsCode = "BE0101N1",
  Tid = years
)
```

The complete table is much larger than a typical interactive API request. Its dimensions contain 312 region codes, four marital-status categories, 102 age categories, two sex categories, two content variables, and 57 years. Downloading the full table would imply more than 29 million cells. Statistics Sweden's PxWeb API version 2 configuration currently limits a single data response to 150,000 data cells. Our subset contains 967,176 cells, which is also above the single-response limit. Calling `pxweb_get()` validates the query, splits it into seven batches, downloads the pieces, and combines them into one `pxweb_data_v2` object.

```
px_data <- pxweb_get(metadata_url, query = query)
```

```
Downloading large query (in 7 batches):
|======================================================================| 100%
```

Comments, notes, and observation statuses can be extracted from the downloaded object before conversion. The same call returns an empty comments data frame for responses without comments.

```
comments <- pxweb_data_comments(px_data)
comments_df <- as.data.frame(comments, stringsAsFactors = FALSE)
comments_df[, c("comment_type", "comment")]
```

```
    comment_type    comment
1    obs_comment    Heby municipality changed county affiliation ...
2 column_comment    The period 1968-1998 is reported according to ...
```

The downloaded object can then be converted directly to a standard `data.frame` for analysis.

```
population <- as.data.frame(
  px_data,
  column.name.type = "code",
  variable.value.type = "code"
)

dim(population)
```

```
[1] 967176      7
```

The downloaded data object can be cited as follows.

```
pxweb_cite(px_data)
```

```
Statistics Sweden (2026). "Population by region, marital status, age
and sex. Year 1968-2024." [Data accessed ... using pxweb R package
1.0.0].
```

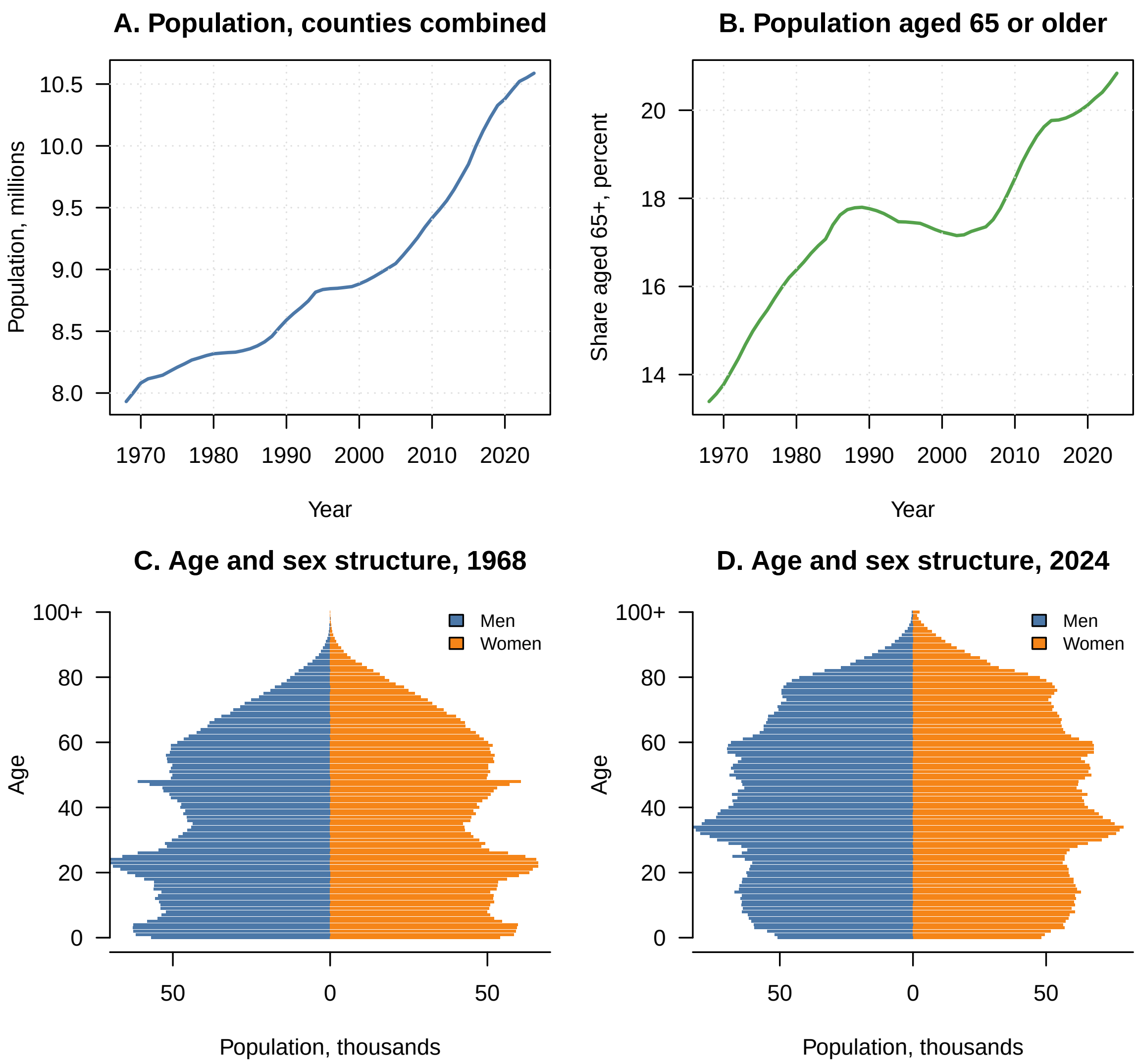

Figure 2: Population development in Sweden based on Statistics Sweden table `TAB638`. The data are downloaded at county level for all years from 1968 to 2024, all single-year age groups from 0 to 100+, both sexes, and all marital-status categories. The request contains 967,176 cells and is split by **pxweb** into seven API calls before being combined for analysis.

Finally, the data are aggregated and plotted. Figure 2 summarizes the downloaded data. The first two panels aggregate counties, marital statuses, ages, and sexes to show the growth of the population and the increasing share of the population aged 65 years or older. The lower panels compare the age and sex structure in 1968 and 2024. The example demonstrates the main package workflow: a PxWeb API version 2 table is found through search, queried programmatically, split because it exceeds the API cell limit, inspected for comments, cited, and transformed into standard R objects for analysis and visualization.

4. Discussion

The **pxweb** package provides access to official statistics shared through PxWeb APIs. It supports the full workflow from API discovery and metadata inspection to query construction, retrieval, conversion, comments, codelists, and citation. In doing so, it helps users move from interactive table browsing to scripted, transparent, and reproducible statistical workflows following the tidy data principles (Wickham *et al.* 2019; Wickham 2014).

The package has matured through active development since its first CRAN release in 2014, including major redesigns in 2018 and 2026 and recent support for PxWeb API version 2. As an indication of use at the time of writing, CRAN download logs report 116,164 downloads between January 2014 and June 2026 (CRAN logs 2026). The package has found direct uses in research, for instance in work on electronic gambling machines and socioeconomic status (Raisamo, Toikka, Selin, and Heiskanen 2019), income-inequality data construction (Solt 2019), trends in youth violence and mortality (Tanskanen, Beuker, Suonpää, Haapakangas, and Kivivuori 2026; Hakkarainen *et al.* 2026), radiation-dose modeling after the Chernobyl nuclear power plant accident (Tondel, Gabrysch, Rääf, and Isaksson 2025), and programmatic access to open statistical data for population studies (Willekens 2023).

Several limitations remain. PxWeb installations differ across providers, provider-side query limits and rate limits vary, and APIs can change independently of the package. These issues motivate the package's emphasis on metadata validation, request splitting, test fixtures, and extensive testing. They also mean that support for PxWeb API version 2 must distinguish between generic API behavior and metadata conventions that are provider-specific.

More broadly, **pxweb** contributes to the open data science ecosystem (Lahti 2018) by making official statistics easier to retrieve, document, and reuse in R. The package does not replace provider documentation or domain knowledge, but it gives researchers, public-sector analysts, educators, and civic users a common programmatic interface to a widely used dissemination infrastructure.

A. Bundled API Catalogue

Acknowledgments

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Bellosta CJG, Viciana FJ, Lamigueiro OP (2025). *pxR: PC-Axis with R.* doi:10.32614/

[1]https://github.ropengov.io

Entry	Provider	Language
`web.dzs.hr`	Croatian Bureau of Statistics	en
`andmed.stat.ee`	Estonia – official statistics	en, et
`statistika.tai.ee`	Estonian Health Statistics and Health Research Database	en
`tilastot.etk.fi`	Finnish Centre for Pensions	en
`tilastot.tela.fi`	Finnish Pension Alliance statistics	en
`trafi2.stat.fi`	Finnish Transport Safety Agency	en, sv, fi
`px.web.ined.fr`	Generations and Gender Contextual Database	en
`pc-axis.geostat.ge`	Geostat Statistics Database	en
`stat.hel.fi`	Helsingin seudun aluesarjat -tilastotietokanta	fi
`pxweb.irena.org`	International Renewable Energy Agency	en
`www6.poderjudicial.es`	Judicial statistics, Spain	es
`data.stat.gov.lv`	Latvia – official statistics	en, lv
`statistika.spkc.gov.lv`	Latvian Health Statistics Database	en
`statistik.linkoping.se`	Linkoping municipality in Sweden	sv
`statdb.luke.fi`	LUKE Natural Resources Institute Finland	en, fi, sv
`pxweb.nhwstat.org`	Nordic Health and Welfare Statistics	en
`pxweb.nordicstatistics.org`	Nordic Statistics Database	en
`openstat.psa.gov.ph`	Philippine Statistics Authority OpenSTAT	en
`www.grande-region.lu`	Portail statistique de la Grande Region	fr, de
`pxweb.stat.si`	SiStat Database	sl
`bank.stat.gl`	Statbank Greenland	en, kl, da
`makstat.stat.gov.mk`	State Statistical Office of the Republic of Macedonia	en, mk
`pxweb.asub.ax`	Statistics Aland	en, sv
`statbank.hagstova.fo`	Statistics Faroe Islands	en, fo
`statfin.stat.fi`	Statistics Finland	en, fi, sv
`px.hagstofa.is`	Statistics Iceland	en, is
`askdata.rks-gov.net`	Statistics Kosovo	en
`etab.llv.li`	Statistics Liechtenstein	en
`statbank.statistica.md`	Statistics Moldova	en, ro
`api.scb.se`	Statistics Sweden	en, sv
`www.pxweb.bfs.admin.ch`	Statistics Switzerland	en, de, fr
`m02-http-pxwebb.login.sundsvall.se`	Sundsvall municipality in Sweden	sv
`statistik.tillvaxtanalys.se`	Swedish Agency for Growth Policy Analysis	sv
`statistik.csn.se`	Swedish Board of Student Finance	sv
`pxexternal.energimyndigheten.se`	Swedish Energy Agency	en
`skogsstatistik.slu.se`	Swedish University of Agricultural Sciences forest statistics	en
`fohm-app.folkhalsomyndigheten.se`	The Public Health Agency of Sweden	sv
`statistik.sjv.se`	The Swedish Agricultural Agency	sv
`pxweb.skogsstyrelsen.se`	The Swedish Forest Agency	en
`statistik.konj.se`	The Swedish National Institute of Economic Research	en, sv
`prognos.konj.se`	Swedish National Institute of Economic Research, forecast database	sv
`w3.unece.org`	United Nations Economic Commission for Europe	en
`statistik.vasteras.se`	Vasteras municipality in Sweden	sv
`pxweb2022.vgregion.se`	Vastra Gotaland Region in Sweden	sv
`vero2.stat.fi`	Verohallinto – Finnish Tax Administration	en, fi, sv
`visitfinland.stat.fi`	Visit Finland (Rudolf service)	fi

Table 3: Bundled PxWeb API catalogue entries in **pxweb**. The table lists the catalogue key, provider description, and available languages.

CRAN.package.pxR. R package version 0.42.8, URL https://CRAN.R-project.org/package=pxR.

Blondel E (2014). *rsdmx: Tools for Reading SDMX Data and Metadata*. R package version 0.6-3, URL https://CRAN.R-project.org/package=rsdmx.

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